



इंदिरा गांधी राष्ट्रीय मुक्त विश्वविद्यालय
मैदान गढ़ी, नई दिल्ली - 110068

Indira Gandhi National Open University
Maidan Garhi, New Delhi - 110068



IGNOU - Student Identity Card

Enrolment Number : 2352311928

RC Code : 07: DELHI 1 (MOHAN ESTATE (SOUTH DELHI))

Name of the Programme : BCA : BACHELOR OF COMPUTER APPLICATIONS

Name : MD.ZEESHAN ANSARI

Guardian's Name : MOHD.SHAHABUDDIN

Address : B-204, Ekta Vihar,, Jaitpur, Badarpur New Delhi
DELHI

Pin Code : 110044



- Instructions :
1. This card should be produced on demand at the Study Center, Examination Center or any other Establishment of IGNOU to use its facilities.
 2. The facilities would be available only relating to the Programme/course for which the student is registered.
 3. This ID Card is generated online. Students are advised to take a color print of this ID Card and get it laminated.
 4. The student details can be cross checked with the QR Code at www.ignou.ac.in. PD 2025-12-29

Registrar
Student Registration Division



INDIRAGANDHINATIONALOPENUNIVERSITY

Regional Centre Delhi-1

Format for Assignment Submission

For Term End Exam June/December- June (Year) 2026

(Please read the instructions given below carefully before submitting assignments)

1. Name of the Student : Mr Zeeshan Ansari
2. Enrollment Number : 2352311928
3. Programme Code : BCA
4. Course Code : BCSL-063
5. (Use this format course-wise separately)
Study Centre Code : 07162(P)
6. Name of the Study Centre : MERIT, A-9, USO Road, Outub
With complete address : Institutional Area, Delhi-110067
7. Mobile Number : 8512881364
8. E-mail ID : zeeshan.ansari012345@gmail.com
9. Details if this same assignment has :
been submitted anywhere else also
10. Above information is cross checked and it is correct: Yes/No

Date of Submission: 9-4-2026

Zeeshan
(Signature of the student)

A. Important Instructions:-

1. Please do not send any assignment at any email of the Regional Centre, it will not be considered.
2. Please avoid duplicacy. Do not re-submit the same assignment anywhere else or by any other means.
3. About the mode of submission of assignments, pl wait for instructions from IGNOU Hqtrs. As soon as we shall come to know, we will share it with all.
4. Please do not use plastic covers. Use plain A4 size pages for assignments for uniformity and better management with this cover page format on each assignment.
5. Please write your name and enrollment no. at the bottom of each page of your assignment.
6. Please retain a photocopy set of assignment submitted with you for record (may be asked to submit at later stage) and also keep the assignment submission receipt in safe custody.
7. If assignment awards are not updated in your Grade Card within next 09 months, please write to us at rcdelhi1@ignou.ac.in giving your complete details and attaching the proof of assignment submission.
8. Assignment Question Paper can be downloaded from: <https://webservices.ignou.ac.in/assignments/>

B. Compulsory sequence of the Assignment Set:

1. Duly Filled in Assignment Submission Cover Page (This Format Page).
2. Copy of IGNOU Identity Card.
3. Print out of valid/applicable assignment question paper.
4. Handwritten Assignment, written on both the sides of page (preferably plain A4 size).

Course Code	:	BCSL-063
Course Title	:	Operating System Networking Management Lab
Assignment Number	:	BCA(VI)/L-063/Assignment/2025-26
Maximum Marks	:	50
Weightage	:	25%
Last date of Submission	:	31st October, 2025 (For July, Session)
	:	30th April, 2026 (For January, Session)

Note: This assignment has five questions for a total of 40 marks. The rest 10 marks are for viva voce. For each question, you must write the required shell script or provide a step-by-step procedure with screenshots. Attach all scripts and screenshots with your assignment.

Q1. Shell Scripting (Linux/UNIX)

(a) Write a shell script named backup.sh that performs the following tasks:

1. It should accept a source directory path and a destination directory path as command-line arguments.
2. It should check if the source directory exists. If not, it should print an error message and exit.
3. It should create a compressed backup of the source directory in the destination directory. The backup filename should be in the format backup_YYYY-MM-DD.tar.gz (e.g., backup_2025-10-25.tar.gz).
4. After a successful backup, it should print a message: "Backup of [source_directory] completed successfully at [destination_path/backup_filename]".

(b) Write another shell script named user_check.sh that takes a username as input. The script should check if the given user is currently logged into the system.

- If the user is logged in, it should display: "[Username] is currently active on the system."
- If the user is not logged in, it should display: "[Username] is not logged in."

(8 Marks)

Q2. Linux/UNIX System Administration

Perform the following system administration tasks. For each task, write the exact command(s) you would use and provide a brief explanation of what the command does.

- (a)** Create a new user account named intern with /home/intern as their home directory and /bin/bash as their default shell.
- (b)** Create a new group named devteam. Add the user intern and an existing user manager to this new group.
- (c)** Create a directory named /project/data. Change its ownership so that the user manager is the owner and the group devteam is the group owner.
- (d)** Set the permissions for the /project/data directory such that:
 - * The owner (manager) has read, write, and execute permissions.
 - * Members of the devteam group have read and execute permissions only.
 - * Other users have no permissions at all.

(8 Marks)

Q3. Windows Server Administration

Provide a step-by-step procedure with screenshots for the following tasks in a Windows Server environment.

(a) Create and Configure a New User:

1. Open "Active Directory Users and Computers".
2. Create a new user with the following details:
 - o Full Name: Riya Sharma
 - o User logon name: riya.s
3. Set a password for the user and configure the account so that the "User must change password at next logon".
4. Add this new user to the "Marketing" group (assume the group already exists).

(b) Create a Shared Folder:

1. Create a new folder named MarketingReports on the C: drive.
2. Share this folder with the network name Marketing_Share.
3. Configure the share permissions so that the "Marketing" group has "Full Control".
4. Configure the NTFS permissions so that the "Marketing" group has "Modify" permissions, and the "Administrators" group has "Full Control".

(8 Marks)

Q4. Network Configuration and Services

Provide a step-by-step procedure with screenshots to perform the following network configurations.

(a) Configure a Static IP Address (Windows):

Configure a network adapter in Windows with the following static IP settings:

- IP Address: 192.168.1.50
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.1
- Preferred DNS Server: 8.8.8.8

(b) Install and Configure a Basic DHCP Server (Windows Server):

1. Install the DHCP Server role on your Windows Server.
2. Create a new IPv4 scope with the following details:
 - o Scope Name: Office_Network
 - o Start IP Address: 192.168.1.100
 - o End IP Address: 192.168.1.200
 - o Subnet Mask: 255.255.255.0
 - o Router (Default Gateway): 192.168.1.1
 - o DNS Server: 8.8.8.8
3. Activate the scope.

(8 Marks)

Q5. Linux/UNIX Command-Line Tasks

Write the single-line command to perform each of the following tasks in a Linux/UNIX terminal.

- (a)** Find and display all lines in the file `/var/log/syslog` that contain the word "error" (case-insensitive).
- (b)** List all running processes on the system and find the Process ID (PID) of the `apache2` process.
- (c)** Create a symbolic link named `webapp` in your home directory that points to `/var/www/html`.
- (d)** Display the disk space usage of all mounted file systems in a human-readable format (e.g., KB, MB, GB).
- (e)** Schedule a job using `cron` to run the script `/home/user/maintenance.sh` at 2:00 AM every Sunday.

(8 Marks)

Q.7 (9)

Solution

#!/bin/bash

check if correct number of argument is provided
if ["\$#" -ne 2]; thenecho "usage: \$0 <source directory> <destination directory>"
exit 1
fi

src_dir="\$1"

dest_dir="\$2"

date_str=\$(date +%Y-%m-%d)

backup_file="backup_\${date_str}.tar.gz"

check if source directory exists

if [! -d "\$src_dir"]; then

echo "error: Source directory '\$src_dir' does exist".

exit 1

fi

create backup

tar -czf "\$dest_dir/\$backup_file" -C "\$src_dir".

check if backup was created

if [\$? -eq 0]; then

echo "Backup of \$src_dir completed successfully
at \$dest_dir/\$backup_file".

else

echo "error: backup failed".

fi.

Tedi?

Explanation

- The script check if two arguments are provided.
- It verifies that the source directory exists.
- The backup filename is generated using the current date.
- The Tar command creates a compressed backup in the destination directory.

Q.7 (b)

Solution

```
# ! bin / bash
# checks if username is provided
if [ "$#" -ne 1 ]; then
    echo "usage: $0 <username>"
    exit 1
fi

username="$1"
# check if user is logged in
if who | grep -w "$username" > /dev/null; then
    echo "$username is currently active on system".
else
    echo "$username is not logged in".
fi
```

Explanation

- The script require a username is a command-line argument.
- if found it print the "Active" message: otherwise it prints "not logged in".

Tuti?

Q.2 (a)

Solution

command:

`Sudo useradd -m -d /home/intern -s /bin/bash intern`Explanation:

This command creates a new user called intern, sets the Home directory to /home/intern, and the default login shell to /bin/bash. The -m option ensures the Home directory is created if it doesn't exist.

Q.2 (b)

Solution`Sudo groupadd devteam``Sudo usermod -g devteam intern``Sudo usermod -g devteam Manager`Explanation

- The first command creates a new group named devteam.
- The next two commands add user intern and manager to the devteam group.

Q.2 (c)

Solution`Sudo mkdir -p /Project/Data``Sudo chown manager:devteam /Project/Data`Explanation

- The first command creates the Project Data directory.
- The second command change the owner of directory to Manager.

Q.2 (1)

Solution

command:

sudo chmod 750 / Project Data.

Explanation

- owner
- group
- others.

Q.3 (9)

Solution

- Step 1: Open Active Directory user and computers
- I click the Start button, Search for Active Directory user and computers and open it.

Screenshot 1: Active Directory user and computers.

Step 2: create A new user

- In the left Panel, I browse to the organizational unit where I want the new user
- I right click the ou, select New > user.

Screenshot 2: selecting 'user > new' in AD user and computers.

Tili

- I enter the following details:
 - Full name: Kiya Bhasma
 - user login name: siya.B
- I click next.

Screenshot: entering user details.

Step 3:

Set Password and Require change.

- I set a Password for Kiya Bhasma.
- I ensure the user must change password at next login box is checked.
- I click next, then finish.

Screenshot 4: Password And Account options

Screen with user must change password at Next.

Q. 3(b)
Solution

Step 1: create folder

- I open file explorer and navigate to C: drive
- I right click inside the drive, select New > Folder, and Name it Marketing Reports

Step 2: Share the Folder

- I right clicked the Marketing Reports folder and Properties.
 - I switch to the sharing tab and click Advanced Sharing
 - I check share this folder and set the share name.
- Then, I click permissions.

Tick?

Step 3: Set Share Permissions

- In the permission window, I remove the default everyone.
- I click add, type marketing click OK.
- I select the marketing group and check full control.
- I click Apply And OK to save the permission.

Step 4: configure NTFS permission

- Back in the properties window, I go to the security tab.
- I click edit then Add
- I add the Marketing group and assign Modify permission.
- finally, I click Apply And OK.

Q.4

Solution

Step 1: open Network connections

- I click the Start button, type NCPA.cpl, And press enter to open Network connection

Screenshot 1: Network connections window showing available Adapters.

Step 2: open Network Adapter Properties.

- I right click the network Adapter I want to configure and select Properties

Screenshot 2: Adapter Properties window.

Step 3: Open IPv4 Properties

- In the list under the networking tab, I select Internet Protocol version 4.
- I click the properties button.

Screenshot 3: IPv4 Properties Window.

Step 4: Enter Static IP configuration

- I select use the following IP Address
 - IP address: 192.168.1.50
 - Subnet mask: 255.255.255.0.
 - Default gateway: 192.168.1.1
- I use the following DNS Server Addresses and enter:
 - Preferred DNS Server: 8.8.8.8

Screenshot 4: Static IP and DNS entered in IPv4.

Step 5: close and verify.

- I click close or OK on all windows.

Q.4(b)
Solution

Step 1: install DHCP Server Role

- I open Server Manager
- I click on manage > Add Role And features
- I select on based on feature based Installation, Next
- I select the server from the pool and click Next
- I check the DHCP Server Role, and click Next.

Step 2: create a New IPv4 Scope.

- In Server manager, I open the DHCP management console from Tools > DHCP.
- I expand the Server Node, right click on IPv4, and select New.
- In the New Scope Wizard
 - Scope Name : office - Network
 - click Next.
- I specify the IP Range.
 - start IP : 192.168.1.100
 - end IP : 192.168.1.200
 - click Next.
- I set the Subnet mask to 255.255.255.0 and click next
- I enter the router as 192.168.1.1 And click Next.
- I enter the DNS server as 8.8.8.8 And click Next.

Step 3: Verify Scope Activation

- I confirm the new Scope office - Network appears under IPv4 and its status is Active.

Q.5 (a)

Solution

command:

```
[ gnet -i "ensn0" | var | log | syslog ]
```

~~this~~

(b)

Solution

command:

```
ps aux | gnet | grep -v gnet | awk '{print $2}'
```

(c)
Solution

command :
[ln -s /var /www /html ~ /webapp]

This command creates a symbolic link named "webapp" in the home directory.

(d)
Solution

command :
[df -h]

This shows disk space usage showing sizes in KB, MB or GB for easier reading.

(e)
Solution

command :
[(crontab -l &> /dev/null; echo "0 2 ** * / home / user / maintenance.sh") | crontab -]

This appends a cron schedule entry that runs the maintenance script at 2:00 AM every Sunday.